Assignment_Day_3

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Date.java | setDate.java

```java
public class Date {

    public static void main(String[] args) {

        setDate d = new setDate();
        int choice;

        do {
            System.out.println("\n--- MENU ---");
            System.out.println("1. Set Date");
            System.out.println("2. Add Days");
            System.out.println("3. Add Months");
            System.out.println("4. Add Year");
            System.out.println("5. Display");
            System.out.println("6. Exit");
            System.out.print("Enter choice: ");

            choice = getNum.getchoice();

            switch(choice) {

            // set date
            case 1:
                System.out.println("Enter the number of day:");
                int day = getNum.getchoice();
                System.out.println("Enter the number of month:");
                int month = getNum.getchoice();
                System.out.println("Enter the year:");
                int year = getNum.getchoice();

                d.setDateFormat(day, month, year);
                d.display();
                break;

            // add days
            case 2:
                System.out.println("Enter number of days to add:");
                int daysToAdd = getNum.getchoice();
                d.addDays(daysToAdd);
                d.display();
                break;
```

```
            break;

        // add months
        case 3:
            System.out.println("Enter number of months to add:");
            int monthsToAdd = getNum.getchoice();
            d.addMonths(monthsToAdd);
            d.display();
            break;

        // add year
        case 4:
            System.out.println("Enter number of years to add:");
            int yearsToAdd = getNum.getchoice();
            d.addYears(yearsToAdd);
            d.display();
            break;

        // Display
        case 5:
            d.display();
            break;

        // Exit
        case 6:
            System.out.println("Exiting...");
            break;

        default:
            System.out.println("Enter the valid choice!!!!");
            break;
        }

    } while(choice != 6);

}
```

setData.java

Date.java | setDate.java

```java
class setDate {

    private int day;
    private int month;
    private int year;

    setDate(int day, int month, int year) {
        setDateFormat(day, month, year);
    }

    setDate() {
        this.day = 1;
        this.month = 1;
        this.year = 2026;
    }

    public void setDateFormat(int day, int month, int year) {

        this.year = (year > 0) ? year : 2026;

        if (month < 1 || month > 12) {
            this.month = 1;
        } else {
            this.month = month;
        }

        int maxDays = getDaysInMonth(this.month, this.year);
        if (day < 1 || day > maxDays) {
            this.day = 1;
        } else {
            this.day = day;
        }
    }

    public boolean leapYear(int year) {
        if (year < 0) {
            return false;
        }

        if (year % 400 == 0 || (year % 100 != 0 && year % 4 == 0)) {
            return true;
        }

```

Date.java | setDate.java

```
46
47      public int getDaysInMonth(int month, int year) {
48          if (month == 1 || month == 3 || month == 5 || month == 7 || month == 8 || month == 10 || month == 12) {
49              return 31;
50          } else if (month == 4 || month == 6 || month == 9 || month == 11) {
51              return 30;
52          } else {
53              return leapYear(year) ? 29 : 28;
54          }
55      }
56
57      public void addDays(int days) {
58          if (days < 0) return;
59          this.day += days;
60          while (this.day > getDaysInMonth(this.month, this.year)) {
61              this.day -= getDaysInMonth(this.month, this.year);
62              this.month++;
63              if (this.month > 12) {
64                  this.month = 1;
65                  this.year++;
66              }
67          }
68      }
69
70      public void addMonths(int months) {
71          if (months < 0) return;
72          this.month += months;
73          while (this.month > 12) {
74              this.month -= 12;
75              this.year++;
76          }
77          int maxDays = getDaysInMonth(this.month, this.year);
78          if (this.day > maxDays) {
79              this.day = maxDays;
80          }
81      }
82
83      public void addYears(int years) {
84          if (years < 0) return;
85          this.year += years;
86          int maxDays = getDaysInMonth(this.month, this.year);
87          if (this.day > maxDays) {
88              this.day = maxDays;
89          }
```

```
81      }
82
83      public void addYears(int years) {
84          if (years < 0) return;
85          this.year += years;
86          int maxDays = getDaysInMonth(this.month, this.year);
87          if (this.day > maxDays) {
88              this.day = maxDays;
89          }
90      }
91
92      public int getDay() {
93          return day;
94      }
95
96      public int getMonth() {
97          return month;
98      }
99
100     public int getYear() {
101         return year;
102     }
103
104     void display() {
105         System.out.println(day + "/" + month + "/" + year);
106     }
107
108 }
```